



Lovely Professional University

Report of INT222

Project Title: Event Management System

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1. INTRODUCTION

1.1 PURPOSE

This Software Requirements Specification (SRS) document outlines the functional and non-functional requirements for "EventFlow," a comprehensive Event Management System built using the MERN stack (MongoDB, Express.js, React.js, Node.js). The intended audience includes software developers, testers, and project stakeholders involved in the development of the system.

1.2 SCOPE

EventFlow is a web-based event management platform that enables:

- User registration, authentication, and profile management
- Event creation, management, and RSVP tracking
- Admin control panel for user and event management
- Organizer role assignment for specific events
- RSVP status tracking and management
- Event approval workflow system

In-Scope:

- Complete user authentication system
- Event creation and management
- RSVP tracking with status updates
- Admin dashboard for system oversight
- Role-based access control (User, Organizer, Admin)
- Email notifications using Nodemailer

Out-of-Scope:

- Mobile application development
- Payment gateway integration
- Video conferencing features
- Cloud deployment requirements
- Third-party authentication services

1.3 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

- **MERN:** MongoDB, Express.js, React.js, Node.js
- **RSVP:** Répondez S'il Vous Plaît (French for "please respond")
- **API:** Application Programming Interface
- **UI/UX:** User Interface/User Experience
- **CRUD:** Create, Read, Update, Delete
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1.4 REFERENCES

1. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
2. MERN Stack Documentation
3. React.js Official Documentation
4. MongoDB Documentation
5. Node.js Documentation

1.5 OVERVIEW

This document is organized into four main sections: Introduction, General Description, Specific Requirements, and Analysis Models. Section 3 contains detailed functional and non-functional requirements. Appendices contain supplementary information including user stories and wireframes.

2. GENERAL DESCRIPTION

2.1 PRODUCT PERSPECTIVE

EventFlow is a standalone web application designed to streamline event management processes. The system operates as a local development project with potential for future deployment.

2.2 PRODUCT FUNCTIONS

1. User registration and authentication
2. Event creation and management
3. RSVP tracking and management
4. Admin approval workflow
5. Organizer role assignment
6. Email notifications using Nodemailer
7. User profile management

2.3 USER CHARACTERISTICS

Primary Users:

1. **Regular Users:** Register for events, track RSVPs, view event details
2. **Event Organizers:** Create and manage specific events, track RSVPs
3. **Administrators:** Manage all users, approve events, assign organizer roles

Technical Proficiency:

- All users: Basic web navigation skills
- Organizers and Admins: Moderate technical understanding

2.4 GENERAL CONSTRAINTS

1. Must be developed using MERN stack
2. Responsive design for desktop and mobile viewing
3. Must use local development environment
4. Nodemailer for email functionality
5. Session-based authentication

2.5 ASSUMPTIONS AND DEPENDENCIES

Assumptions:

1. Development environment with Node.js and MongoDB installed
2. Local email server or test email configuration
3. Modern web browsers for testing

Dependencies:

1. Node.js runtime environment
2. MongoDB database

3. Nodemailer package for email functionality
 4. Express-session for session management
-

SPECIFIC REQUIREMENTS

2.6 EXTERNAL INTERFACE REQUIREMENTS

2.6.1 User Interfaces

- **Login/Registration Page:** Clean form with validation
- **Dashboard:** Role-specific views with event summaries
- **Event Creation Form:** Multi-step form with category selection
- **RSVP Management Interface:** Visual status tracking
- **Admin Panel:** Comprehensive management interface
- **Responsive Design:** Works on desktop and mobile browsers

2.6.2 Hardware Interfaces

- Standard development machine requirements
- Minimum: 8GB RAM, modern processor for development

2.6.3 Software Interfaces

- **Frontend:** React.js with state management
- **Backend:** Node.js with Express.js framework
- **Database:** MongoDB with Mongoose ODM
- **Authentication:** Session-based with password hashing

2.6.4 Communications Interfaces

- RESTful API endpoints
- Nodemailer for email communications
- Local server communication

2.7 FUNCTIONAL REQUIREMENTS

2.7.1 User Authentication and Authorization

FR-001: User Registration

- **Inputs:** Email, password, name, phone number
- **Processing:** Email validation, password hashing, duplicate check
- **Outputs:** User account creation, success message
- **Error Handling:** Duplicate email, invalid format, weak password

FR-002: User Login

- **Inputs:** Email and password
- **Processing:** Credential validation, session creation
- **Outputs:** Session establishment, user role identification
- **Error Handling:** Invalid credentials

FR-003: Role-Based Access Control

- **Introduction:** Three-tier role system (User, Organizer, Admin)
- **Processing:** Session validation, middleware protection
- **Outputs:** Role-specific interface rendering

FR-004: User Profile Management

- **Inputs:** Profile updates (name, phone, preferences)
- **Processing:** Data validation, database update
- **Outputs:** Updated profile information
- **Error Handling:** Invalid data, unauthorized access

2.7.2 Event Management

FR-005: Event Creation

- **Inputs:** Event title, description, date, time, location, capacity, category
- **Processing:** Data validation, organizer assignment, status setting
- **Outputs:** Event record creation, admin notification email
- **Error Handling:** Invalid dates, missing required fields, capacity errors

FR-006: Event Approval Workflow

- **Introduction:** Admin approval required for all events
- **Processing:** Admin review, approval/rejection with comments
- **Outputs:** Event status update, email notification to organizer
- **Error Handling:** Missing admin approval, invalid status transitions

FR-007: Event Browsing and Search

- **Inputs:** Search terms, filters (date, category, status)
- **Processing:** Database query with filters
- **Outputs:** Filtered event list, event details
- **Error Handling:** No results found, invalid filter criteria

FR-008: Event Updates and Cancellation

- **Inputs:** Updated event details, cancellation reason
- **Processing:** Validation, status update, notification
- **Outputs:** Updated event record, attendee notifications
- **Error Handling:** Unauthorized edits, event in progress

2.7.3 RSVP Tracking System

FR-009: RSVP Submission

- **Inputs:** User selection (Attending/Not Attending/Maybe)
- **Processing:** Check event capacity, update attendance count
- **Outputs:** RSVP status update, confirmation email
- **Error Handling:** Event at capacity, duplicate RSVP, event closed

FR-010: RSVP Dashboard for Organizers

- **Introduction:** Visual representation of attendance statistics
- **Processing:** Real-time data aggregation from database
- **Outputs:** Attendance charts, statistics, attendee lists
- **Error Handling:** No attendees, data loading errors

FR-011: RSVP Status Management

- **Inputs:** RSVP status changes by user
- **Processing:** Update attendance records, recalculate statistics
- **Outputs:** Updated RSVP status, organizer notification
- **Error Handling:** Invalid status change, event passed

FR-012: Attendance Tracking

- **Introduction:** Track actual attendance vs RSVPs
- **Processing:** Check-in system, comparison with RSVPs

Project Name

- **Outputs:** Attendance reports, no-show lists
- **Error Handling:** Duplicate check-ins, invalid user

2.7.4 Admin Dashboard

FR-013: User Management

- **Inputs:** User search, filter criteria, role changes
- **Processing:** User data retrieval, role modification, account management
- **Outputs:** User list, editable profiles, activity logs
- **Error Handling:** Invalid user operations, permission conflicts

FR-014: Event Oversight and Approval

- **Introduction:** Comprehensive event monitoring and approval queue
- **Processing:** Event status tracking, bulk operations, analytics
- **Outputs:** Event analytics, approval queues, system reports
- **Error Handling:** Approval conflicts, data inconsistencies

FR-015: System Configuration

- **Inputs:** System settings, email templates, notification preferences
- **Processing:** Configuration validation, database updates
- **Outputs:** Updated system settings, confirmation messages
- **Error Handling:** Invalid configurations, save failures

2.7.5 Organizer Management

FR-016: Organizer Assignment

- **Inputs:** User selection, event assignment, permission level
- **Processing:** Role elevation, permission assignment, validation
- **Outputs:** Organizer privileges, notification emails
- **Error Handling:** User not found, insufficient permissions, duplicate assignment

FR-017: Organizer Dashboard

- **Introduction:** Event-specific management interface with RSVP tracking
- **Processing:** RSVP tracking, attendee management, event updates
- **Outputs:** Event statistics, attendee lists, management tools
- **Error Handling:** No events assigned, permission issues

FR-018: Event-Specific Organizer Controls

- **Inputs:** Event modifications, attendee management actions
- **Processing:** Permission checks, data validation, updates
- **Outputs:** Updated event data, attendee notifications
- **Error Handling:** Unauthorized actions, invalid data

2.7.6 Notification System

FR-019: Email Notifications via Nodemailer

- **Inputs:** Event triggers (RSVP, approval, updates, reminders)
- **Processing:** Template selection, recipient identification, email composition
- **Outputs:** Email delivery via Nodemailer, delivery status logging
- **Error Handling:** Email service errors, invalid recipients, template errors

FR-020: Notification Types

- **Introduction:** Various notification types for different events
- **Processing:** Event detection, appropriate notification selection
- **Outputs:** Context-specific emails to relevant users

- **Error Handling:** Missing templates, user preferences not respected

FR-021: Notification Preferences

- **Inputs:** User preference settings for notification types
- **Processing:** Preference storage, filtering based on preferences
- **Outputs:** Customized notification delivery
- **Error Handling:** Invalid preferences, save failures

2.8 NON-FUNCTIONAL REQUIREMENTS

2.8.1 Performance

- Page load time < 3 seconds on local development environment
- API response time < 500ms for database operations
- Support at least 50 concurrent users in development testing
- Database queries optimized with proper indexing
- Efficient image loading and caching strategies

2.8.2 Reliability

- Proper error handling for all user interactions
- Data consistency maintained across all operations
- Session management reliable with proper timeouts
- Email notification system with retry mechanism
- Database backup procedures for development data

2.8.3 Availability

- System accessible during development hours
- Graceful degradation when services are unavailable
- Clear maintenance notifications if needed
- Local server stability for demonstration purposes

2.8.4 Security

- Password hashing using bcrypt with appropriate salt rounds
- Session security with proper expiration and regeneration
- Input validation and sanitization for all user inputs
- SQL injection prevention through parameterized queries
- XSS protection through output encoding
- CSRF protection for state-changing operations
- Secure email configuration for Nodemailer
- Environment variables for sensitive configuration

2.8.5 Maintainability

- Modular code structure with clear separation of concerns
- Comprehensive code documentation and comments
- Consistent coding standards and naming conventions
- Unit test coverage for critical components
- API documentation for backend endpoints
- Version control using Git with meaningful commit messages
- Clear project structure and configuration files

2.8.6 Portability

- Cross-browser compatibility (Chrome, Firefox, Safari, Edge)
- Responsive design for various screen sizes
- Environment-agnostic configuration

Project Name

- Easy setup with detailed documentation
- Docker support for consistent development environment (optional)

2.9 DESIGN CONSTRAINTS

1. **Technology Stack:** Must use MERN stack exclusively
2. **Database:** MongoDB with Mongoose ODM
3. **Authentication:** Session-based with express-session
4. **Email Service:** Nodemailer for all email functionality
5. **State Management:** React state hooks and Context API
6. **Styling:** CSS modules or styled-components
7. **API Design:** RESTful principles
8. **Code Standards:** ESLint configuration, Prettier formatting
9. **Testing:** Jest for backend, React Testing Library for frontend
10. **Development Environment:** Local development only

2.10 OTHER REQUIREMENTS

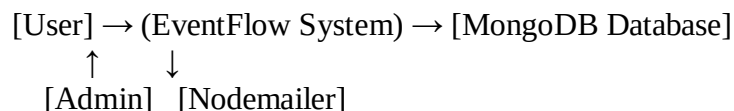
1. **Development Setup:** Clear setup instructions for other developers
2. **Testing Data:** Scripts to populate database with test data
3. **Documentation:** README with setup and usage instructions
4. **Error Logging:** Console logging for development debugging
5. **Code Quality:** Pre-commit hooks for code formatting
6. **Dependency Management:** Regular updates and security checks

3. ANALYSIS MODELS

3.1 DATA FLOW DIAGRAMS (DFD)

Context Level DFD:

text



Level 1 DFD Processes:

1. User Authentication Flow
2. Event Creation and Approval Flow
3. RSVP Processing and Tracking Flow
4. Email Notification Dispatch Flow
5. Admin Management Flow

3.2 USE CASE DIAGRAM

Actors:

1. **User:** Register, Login, Browse Events, RSVP, Manage Profile
2. **Organizer:** Create Events, Manage RSVPs, Update Event Details
3. **Admin:** Manage Users, Approve Events, Assign Roles, System Configuration

Key Use Cases:

Project Name

- UC-001: Register New Account
- UC-002: Login to System
- UC-003: Create Event Request
- UC-004: Approve/Reject Event
- UC-005: Submit RSVP
- UC-006: Track RSVP Status
- UC-007: Assign Organizer Role
- UC-008: Send Event Notifications
- UC-009: Configure Email Settings

3.3 ENTITY-RELATIONSHIP DIAGRAM

Entities:

1. User

- user_id (Primary Key)
- email (Unique)
- password_hash
- name
- phone
- role (User/Organizer/Admin)
- created_at
- last_login

2. Event

- event_id (Primary Key)
- title
- description
- date_time
- location
- capacity
- status (Pending/Approved/Rejected/Cancelled)
- organizer_id (Foreign Key)
- created_at
- approved_by (Foreign Key)
- approval_date

3. RSVP

- rsvp_id (Primary Key)
- user_id (Foreign Key)
- event_id (Foreign Key)
- status (Attending/Not Attending/Maybe)
- responded_at
- attended (Boolean)

4. GITHUB LINK

Repository: <https://github.com/tanishqsoni38/EVENT-TRACKER>

5. VIDEO LINK

LinkedIn Link: [[linkedIn link](#)]

A. APPENDICES

A.1 APPENDIX 1: USER STORIES

Priority: High

US-001: User Registration

As a new user, I want to register for an account so I can access the event management system.

US-002: Event Browsing

As a user, I want to browse available events so I can find activities that interest me.

US-003: RSVP Management

As a user, I want to RSVP to events so organizers know my attendance plans.

US-004: Event Creation

As an organizer, I want to create events so I can organize activities for users.

US-005: Event Approval

As an admin, I want to approve or reject events so I can control event quality.

Priority: Medium

US-006: Profile Management

As a user, I want to update my profile information so my details are current.

US-007: RSVP Tracking

As an organizer, I want to track RSVPs so I know how many people are attending.

US-008: User Management

As an admin, I want to manage user accounts so I can maintain system security.

US-009: Email Notifications

As a user, I want to receive email notifications so I stay informed about events.

Priority: Low

US-010: Attendance Reports

As an organizer, I want to generate attendance reports so I can analyze event success.

US-011: Bulk Operations

As an admin, I want to perform bulk operations so I can manage the system efficiently.

A.2 APPENDIX 2: WIREFRAMES

1. **Login/Registration Screen**
 - Email and password fields
 - Register link for new users
 - Remember me option
2. **User Dashboard**
 - Upcoming events section
 - Your RSVPs status
 - Event recommendations
 - Profile quick actions
3. **Event Creation Form**
 - Multi-step form
 - Date/time picker
 - Location input with map (optional)
 - Capacity settings
 - Category selection
4. **Event Details Page**
 - Event information display
 - RSVP buttons
 - Organizer details
 - Attendee list (for organizers)
5. **Admin Dashboard**
 - User management table
 - Event approval queue
 - System statistics
 - Configuration options
6. **Organizer Dashboard**
 - My Events list
 - RSVP statistics
 - Attendee management
 - Event editing tools
7. **RSVP Management Interface**
 - Attendee list with status